STANDATA variance 23-BCV-006

Building

Occupancy classification of natural gas regulating and metering buildings

Date Issued: May 2024

Purpose

To provide an alternative solution for occupancy classification of Group F, Division 1 high-hazard industrial occupancies that house natural gas and metering equipment to be classified as Group F, Division 2 medium-hazard occupancy under certain conditions.

Discussion

High Hazard Industrial Occupancy (Group F, Division 1) is defined by the National Building Code - 2023 Alberta Edition (NBC(AE)) as an industrial occupancy containing sufficient quantities of highly combustible, flammable or explosive materials which, because of their inherent characteristics, may constitute a special fire hazard.

Buildings containing natural gas regulating and metering equipment are typically designed to protect the equipment. These buildings are not accessible by the public and are only intended for occupancy by employees for maintenance so the risk of harm to the public is lower than other buildings. Such features allow for accepting Group F, Division 2 medium-hazard occupancy as an alternative occupancy classification.

Code References

Sentence 6.3.1.5.(1)

6.3.1.5. Indoor Air Contaminants
(See Note A-6.3.1.5.)

1) Air contaminants of concern that are released within *buildings* shall
a) be removed insofar as is possible at their points of origin, and
b) not be permitted to accumulate in concentrations greater than those permitted by applicable provincial or territorial requirements or, in the absence of such requirements, by good engineering practice such as that described in the publications listed in Sentence 6.2.1.1.(1), measured using the methodology described therein.

Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms

High-hazard industrial occupancy (Group F, Division 1) means an *industrial occupancy* containing sufficient quantities of highly *combustible* and flammable or explosive materials which, because of their inherent characteristics, constitute a special fire hazard.

Medium-hazard industrial occupancy (Group F, Division 2) means an *industrial occupancy* in which the *combustible* content is more than 50 kg/m2 or 1 200 MJ/m2 of *floor area* and not classified as a *high-hazard industrial occupancy*.

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Note A-3.7.2.1.(6)

A-3.7.2.1.(6) Low Human Occupancy. Examples of buildings that could be classed as low human occupancy may include pump houses, equipment storage shacks, etc.

Application

This variance applies to Group F, Division 1 high-hazard industrial occupancies housing natural gas and metering equipment.

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

Buildings used to house natural gas regulating and metering equipment may be classified as Group F, Division 2 medium-hazard occupancy provided:

1. the building does not exceed 55 m^2 in building area or one storey in building height,
2. the building is considered as a low human occupancy building, and
3. sufficient ventilation is available as per Sentence 6.3.1.5.(1) so any small leaks from fittings will not accumulate and there is a constant movement of air.

This VARIANCE is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

Issued by the Provincial Building Administrator

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